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Evaluating the role of toe grip strength and anthropometric measures in diagnosing sarcopenia: A predictive model for middle-aged and older adults in Taiwan

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ABSTRACT

Background: Sarcopenia, characterized by the loss of muscle mass and strength, is a major health concern among older adults, contributing to disability and mortality. Early identification is crucial for timely intervention. This study evaluated the role of toe grip strength (TGS) for diagnosing sarcopenia and developed a simplified diagnostic algorithm that reduces reliance on tools such as bioelectrical impedance analysis and handgrip strength. **Methods:** A total of 771 participants (591 women, 180 men), aged 50 and above, were enrolled from a health management center in Taiwan. Participants underwent measurements of calf circumference, handgrip strength, TGS, and body composition. Sarcopenia was defined by the EWGSOP2 criteria: low muscle strength and reduced appendicular skeletal muscle mass index. Logistic regression models assessed the predictive power of age, TGS, and anthropometric measurements for sarcopenia diagnosis.

Results: Sarcopenia was identified in 18.3 % of participants, with a higher percentage in women (18.8 %) than in men (16.7 %). In women, age, height, and calf circumference were strong predictors of sarcopenia, and when combined, resulted in a high AUROC of 0.885. TGS, used alone, had a moderate AUROC of 0.736 and did not significantly improve discrimination when added to the model. In men, TGS was a stronger predictor (AUROC of 0.825), but adding it to the model did not significantly improve diagnostic performance.

Conclusion: Age, height, and calf circumference are effective in identifying sarcopenia in women. TGS may provide insights into mobility but does not enhance sarcopenia diagnosis. Future research should validate these findings and explore novel diagnostic tools.

1. Introduction

Sarcopenia, characterized by the progressive loss of skeletal muscle mass, strength, and function, is a significant health concern among older adults [1]. It is associated with increased risks of falls, fractures, physical disability, and mortality, contributing to a decline in quality of life and imposing a substantial economic burden on healthcare systems globally

[2,3]. Early detection and intervention are crucial for mitigating these adverse outcomes [4].

The definition and diagnostic criteria for sarcopenia have undergone significant evolution, reflecting advances in understanding its pathophysiology and clinical relevance [5]. The European Working Group on Sarcopenia in Older People (EWGSOP) initially defined sarcopenia in 2010, emphasizing low skeletal muscle mass as the core diagnostic

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criterion, supplemented by measures of muscle strength or physical performance [6]. However, accumulating evidence highlighting the predictive value of muscle strength for adverse outcomes led to the revised EWGSOP guidelines in 2019 [7]. These updated criteria (EWGSOP2) prioritize low muscle strength as the primary diagnostic feature, with reduced skeletal muscle mass and impaired physical performance used for confirmation and staging. Concurrently, the Asian Working Group for Sarcopenia (AWGS) updated its consensus in 2019 to address the specific needs of Asian populations [8]. The AWGS 2019 guidelines adapt thresholds and methodologies to reflect regional differences in body composition, lifestyle, and resource availability, ensuring greater relevance to Asian demographics. Despite these shared objectives, the application of these guidelines varies across populations, underscoring the need for diagnostic strategies tailored to specific contexts and available resources.

Toe grip strength (TGS), a novel measure of lower limb function, has been proposed as a potential marker for sarcopenia and is receiving growing research interest. It reflects the strength and functionality of intrinsic foot muscles, which are critical for balance and mobility. Reference values for TGS have been established across age groups, providing a normative basis for its use in functional assessments [9]. While the reliability and reproducibility of the toe grip dynamometer have been validated, its role in sarcopenia diagnosis remains exploratory [10]. Studies have demonstrated significant correlations between TGS and functional outcomes, such as dynamic balance and mobility, highlighting its potential utility in assessing lower limb function [11]. Additionally, TGS has been associated with foot-related parameters such as hallux valgus and foot arch height, further emphasizing its relevance to physical performance and structural integrity [12]. However, its clinical application in sarcopenia assessment remains preliminary, particularly when compared to established measures like handgrip strength and bioelectrical impedance analysis (BIA).

This study focused on adults aged 50 and above in Taiwan who participated in a sarcopenia screening program with comprehensive assessments of muscle mass and strength. It aimed to evaluate the role of TGS in diagnosing sarcopenia and to develop a simplified diagnostic algorithm to reduce reliance on complex tools such as BIA or handgrip dynamometers. These objectives sought to enhance diagnostic accessibility and efficiency for broader clinical application.

2. Methods

2.1. Setting and participants

National Taiwan University Hospital Yunlin Branch (NTUHYL), located in central Taiwan, operates a Health Management Center (HMC) as a self-funded unit providing health examinations for paying clients. These examinations include packages aimed at the prevention of cardiovascular diseases, cancer, and other chronic conditions. Serving over 250 customers per month, the HMC offers comprehensive health checkups tailored to individual needs.

On September 1, 2021, the HMC launched the Sarcopenia Screening Initiative, a research program that employed convenience sampling among its clients. Individuals aged 50 years and above who agreed to participate were included in the study. Exclusion criteria included mobility impairments, significant lower limb functional disabilities, or refusal to participate. The program was approved by the Institutional Review Board (IRB) of National Taiwan University Hospital (IRB #202109082RIND), and all participants provided written informed consent.

2.2. Experiment protocols

Participants first provided basic demographic information, including age and dominant side (right or left hand and foot). Anthropometric measurements, including height and weight, were recorded to calculate

body mass index (BMI).

The calf circumference of the dominant leg was measured using a non-elastic, flexible measuring tape. The measurement was taken at the widest part of the calf, identified by wrapping the tape horizontally around the leg while the participant was seated with their knees bent at a 90-degree angle. The tape was positioned comfortably against the skin without compressing the underlying tissue. A single measurement of the dominant leg was considered sufficient for an accurate assessment.

2.3. Body composition analysis

Body composition was assessed using the ACCUNIQ BC380 Plus (SELVAS Healthcare Co., Ltd., Daejeon, South Korea), which employs bioelectrical impedance analysis (BIA) to measure muscle mass. This technique passes a small, painless electrical current through the body to measure resistance and reactance, which vary depending on the water content and conductivity of different tissues, such as muscle and fat [13].

Before the measurement, participants' palms and soles were cleaned with electrolyte wipes to ensure proper contact with the device's electrodes. Their information, including a unique identification number, height, age, and sex, was entered into the device, which automatically recorded body weight. They stood upright with feet firmly placed on the electrodes and arms positioned slightly away from the torso at approximately a 30-degree angle, remaining still and silent during the 2-min measurement [14].

Segmental muscle mass was measured in all four extremities. Appendicular skeletal muscle mass (ASM) was calculated by summing the muscle mass of both arms and legs, and the appendicular skeletal muscle mass index (ASMI) was derived by dividing ASM by the square of the participant's height (kg/m^2).

2.4. Handgrip strength measurement

Handgrip strength (HGS) was measured using the JAMAR hydraulic handgrip dynamometer (Patterson Medical Supply Inc., Wisconsin, USA) [15]. For the measurement, participants used their dominant hand while seated comfortably with their elbow bent at a 90-degree angle, forearm resting on a flat surface, and wrist in a neutral position. They squeezed the dynamometer with maximum effort for about 3 s. The test was performed three times, with at least 30 s of rest between trials to prevent fatigue. The average value of the three trials was used for analysis.

2.5. Toe grip strength measurement

Toe grip strength was measured using a T.K.K.3362 toe grip dynamometer (Takei Scientific Instruments, Niigata, Japan) [9]. Participants practiced the testing procedure using submaximal effort before the actual measurement to ensure familiarity. The dominant lower extremity, identified as the participant's preferred leg for kicking a ball, was used for the test. Participants were asked to sit comfortably with their knees bent at 90° and their foot placed flat on dynamometer platform. They were instructed to grip the device with their toes using maximum effort for approximately 3 s. The test was performed three times, with a 30-s interval between trials to prevent fatigue. The highest value among the three trials was used for analysis.

2.6. Defining sarcopenia

Sarcopenia was defined based on the criteria established by EWGSOP2. Participants were classified as having sarcopenia if they met both of the following conditions.

1. Low muscle strength, defined as handgrip strength <28 kg for men and <18 kg for women.

2. Reduced skeletal muscle mass, defined as ASMI $\leq 7.0 \text{ kg/m}^2$ for men and $\leq 5.7 \text{ kg/m}^2$ for women, measured using BIA.

The proportion of individuals meeting the diagnostic criteria was analyzed and reported separately for men and women to highlight potential sex-specific differences [7].

2.7. Data analysis

The observed rate of sarcopenia in this cohort was calculated based on the defined criteria of low muscle strength and reduced ASMI. The proportion of participants meeting both criteria was expressed as a percentage. Descriptive statistics summarized the demographic and anthropometric characteristics of the study population, including age, height, weight, BMI, calf circumference, HGS, TGS, and ASMI. Continuous variables were reported as mean $\pm$ standard deviation (SD), while categorical variables were presented as percentages. Pearson correlation coefficients were calculated to assess the relationships between these variables. All analyses were stratified by sex, reflecting sex-specific diagnostic criteria and avoiding the need for complex interaction terms in modeling.

In addition to stratification by sex, participants were further categorized into two age groups: 50–65 years and 65 years and above. This sub-stratification reflects the clinical consideration that the pathophysiology and burden of sarcopenia may differ between middle-aged and older adults. Descriptive statistics, including the proportion of individuals meeting sarcopenia criteria and anthropometric measurements, were summarized for each subgroup. In addition, the lowest quintile values for key variables (calf circumference, TGS, HGS, and ASMI) were calculated separately by age and sex to provide population-specific reference points for future research.

To predict the presence of sarcopenia, three logistic regression models were developed and evaluated using receiver operating characteristic (ROC) curves. Model 1 assessed the predictive value of TGS alone, generating Risk Score 1. Model 2 included demographic and anthropometric predictors (age, height, weight, BMI, and calf circumference) and applied a forward stepwise selection procedure. Variables with $p < 0.05$ were retained, and those with $p > 0.10$ were excluded after iterative re-evaluation, ultimately generating Risk Score 2 based on statistically significant predictors. Model 3 expanded upon Model 2 by incorporating TGS as an additional predictor, resulting in Risk Score 3.

Variance Inflation Factor (VIF) and Tolerance values were calculated to assess multicollinearity among key predictor variables in Model 3. Each variable was alternately treated as the dependent variable in a series of multiple linear regression models, with the remaining variables entered as predictors. This analysis was also conducted separately for women and men.

ROC curves for Risk Scores 1, 2, and 3 were plotted against sarcopenia status, with the area under the ROC curve (AUROC) used to evaluate each model's discriminatory power. Comparisons between the models were conducted using the DeLong test [16]. The results compared Model 1 and Model 2 to assess differences in discriminatory power between TGS alone and the combination of demographic and anthropometric predictors in Model 2. Additionally, the comparison between Model 2 and Model 3 evaluated whether incorporating TGS improved discriminatory power beyond that provided by demographic and anthropometric predictors.

Statistical analyses were conducted using IBM SPSS Statistics version 22 (IBM Corp., Armonk, NY, USA), while the DeLong method was performed in R version 4.3.0 with the *pROC* package. Graphical visualizations, including ROC curves and color contour plots, were generated using Python (version 3.9). To ensure data confidentiality, all participant information was anonymized prior to analysis.

3. Results

From September 1, 2021 to August 31, 2022, a total of 771 individuals aged 50 years and older participated in sarcopenia screening program at the Health Management Center of National Taiwan University Hospital Yunlin Branch. The cohort consisted of 591 women (mean age: 71.4 ± 10.2 years) and 180 men (mean age: 69.0 ± 10.9 years).

Low muscle mass was detected in 36.8 % (284/771) of participants, affecting 39.1 % (231/591) of women and 29.4 % (53/180) of men. Similarly, low HGS was observed in 33.3 % (257/771) of the cohort, with 33.8 % (200/591) of women and 31.7 % (57/180) of men. The frequency of sarcopenia, which requires meeting both low muscle mass and low handgrip strength criteria, was 18.3 % (141/771). Sarcopenia was slightly more common among women (18.8 %, 111/591) compared to men (16.7 %, 30/180). Women in the cohort had lower body weight, height, and muscle strength compared to men. On average, men demonstrated higher HGS, TGS, and ASMI values, reflecting differences in muscle composition and strength characteristics (Table 1, upper panel).

The analysis showed significant relationships between calf circumference, ASMI, and HGS (Table 2). In women, calf circumference positively correlated with ASMI ($r = 0.692$, $p < 0.001$) and HGS ($r = 0.360$, $p < 0.001$). In men, similar correlations were observed between calf circumference and ASMI ($r = 0.781$, $p < 0.001$) as well as HGS ($r = 0.404$, $p < 0.001$). In addition, height was moderately correlated with ASMI in both sexes (women: $r = 0.596$, $p < 0.001$; men: $r = 0.462$, $p < 0.001$).

TGS showed weaker correlations with ASMI and HGS compared to calf circumference (Table 2). In women, TGS correlated with ASMI ($r = 0.321$, $p < 0.001$) and HGS ($r = 0.528$, $p < 0.001$). In men, TGS correlated with ASMI ($r = 0.490$, $p < 0.001$) and HGS ($r = 0.601$, $p < 0.001$).

The results of the predictive model for women (Table 3, upper) revealed that TGS was inversely associated with sarcopenia (odds ratio [OR] = 0.761, $p < 0.001$). However, when used as a standalone predictor in Model 1, its ability to discriminate sarcopenia was moderate, with an AUROC of 0.736. The forward stepwise procedure selected age, height, and calf circumference for inclusion in Model 2, which significantly enhanced its discriminatory power (AUROC = 0.885). This performance was notably superior to that of TGS alone (DeLong test: $p < 0.001$). Model 3, which incorporated TGS alongside the variables from Model 2, did not further improve the model's discrimination, yielding an AUROC of 0.886 (Fig. 1, left). There was no statistically significant difference in AUROC between Model 2 and Model 3 (DeLong test, $p = 0.771$). Based on Model 2, the probability of having sarcopenia at different ages can be visualized using a color contour plot, which highlights the intersection of calf circumference and height (Fig. 2).

For men (Table 3, lower), TGS was inversely associated with sarcopenia in Model 1 (OR = 0.726, $p < 0.001$), achieving an AUROC of 0.825. Model 2, which incorporated age, height, and calf circumference, resulted in an AUROC of 0.903, which was better than that of Model 1 using TGS alone (DeLong test: $p = 0.036$). In this model, age was only borderline significant (OR = 1.062, $p = 0.057$), unlike in women. In Model 3, TGS became a significant predictor (OR = 0.783, $p = 0.006$), but the AUROC showed no significant improvement over Model 2 (AUROC = 0.924, DeLong test: $p = 0.133$) (Fig. 1, right). These results suggest that in men, TGS is a more important predictor than age, but combining the two does not enhance discriminatory power due to shared predictive variance.

3.1. Multicollinearity diagnostics

Multicollinearity diagnostics showed that all VIF values were below 2.0 and Tolerance values above 0.5 across sex-stratified models, indicating low multicollinearity among age, height, calf circumference, and

Table 1

Upper panel: Descriptive Statistics by Sex and Age (Mean $\pm$ Standard Deviation); Lower panel: Cutoff Points for Measurements defined as the lowest quintile (20th percentile).

Descriptive statistics	Women			Men		
	All	65 years old and above	50–65 years old	All	65 years old and above	50–65 years old
n	591	433	158	180	110	70
Age	71.43 $\pm$ 10.22	76.23 $\pm$ 7.03	58.30 $\pm$ 4.38	69.00 $\pm$ 10.89	76.09 $\pm$ 7.09	57.85 $\pm$ 4.66
Weight (kg)	57.84 $\pm$ 9.39	56.86 $\pm$ 9.01	60.53 $\pm$ 9.90	69.97 $\pm$ 11.40	67.40 $\pm$ 10.37	74.02 $\pm$ 11.83
Height (cm)	153.13 $\pm$ 6.06	152.04 $\pm$ 5.81	156.11 $\pm$ 5.72	166.68 $\pm$ 5.76	165.01 $\pm$ 5.38	169.30 $\pm$ 5.38
Body Mass index (BMI)	24.66 $\pm$ 3.74	24.57 $\pm$ 3.51	24.89 $\pm$ 4.30	25.15 $\pm$ 3.64	24.73 $\pm$ 3.38	25.82 $\pm$ 3.94
Calf circumference (cm)	33.02 $\pm$ 3.36	32.40 $\pm$ 3.19	34.72 $\pm$ 3.24	35.45 $\pm$ 3.89	34.17 $\pm$ 3.60	37.45 $\pm$ 3.47
Toe grip strength (kg)	6.32 $\pm$ 4.06	5.25 $\pm$ 3.27	9.27 $\pm$ 4.53	10.65 $\pm$ 6.27	8.09 $\pm$ 5.27	14.67 $\pm$ 5.60
Handgrip strength (kg)	19.89 $\pm$ 5.46	18.61 $\pm$ 5.08	23.38 $\pm$ 4.93	32.68 $\pm$ 8.95	29.03 $\pm$ 7.77	38.41 $\pm$ 7.60
ASMI (kg/m ²)	5.88 $\pm$ 0.79	5.74 $\pm$ 0.76	6.25 $\pm$ 0.74	7.50 $\pm$ 0.84	7.22 $\pm$ 0.80	7.93 $\pm$ 0.70
Sarcopenia (n, %)	111 (18.8 %)	105 (24.2 %)	6 (3.8 %)	30 (16.7 %)	29 (26.4 %)	1 (1.4 %)
The lowest quintile	Women			Men		
	All	65 years old and above	50–65 years old	All	65 years old and above	50–65 years old
n	591	433	158	180	110	70
Calf circumference (cm)	30.00	29.50	32.00	32.00	31.00	35.00
Toe grip strength (kg)	2.94	2.68	5.68	5.10	4.14	9.70
Handgrip strength (kg)	15.30	14.30	19.50	24.36	22.04	32.02
ASMI (kg/m ²)	5.23	5.12	5.68	6.75	6.51	7.32

Table 2

Pearson correlation coefficients among various anthropometric and strength-related variables, * $p < 0.05$, ** $p < 0.001$. The upper panel presents data for women; the lower panel presents data for men. Abbreviations: BMI, body mass index; ASMI, appendicular skeletal muscle mass index.

Women (N = 591)								
	Age	Weight	Height	BMI	Calf circumference	Toe grip strength	Handgrip strength	ASMI
Age	1	−0.238**	−0.381**	−0.067	−0.445**	−0.559**	−0.538**	−0.407**
Weight		1	0.397**	0.876**	0.756**	0.119*	0.238**	0.728**
Height			1	−0.088*	0.335**	0.311**	0.395**	0.596**
BMI				1	0.650**	−0.028	0.056	0.480**
Calf circumference					1	0.338**	0.360**	0.692**
Toe grip strength						1	0.528**	0.321**
Handgrip strength							1	0.423**
ASMI								1
Men (N = 180)								
	Age	Weight	Height	BMI	Calf circumference	Toe grip strength	Handgrip strength	ASMI
Age	1	−0.344**	−0.459**	−0.166*	−0.493**	−0.591**	−0.630**	−0.473**
Weight		1	0.447**	0.903**	0.788**	0.312**	0.271**	0.787**
Height			1	0.023	0.369**	0.369**	0.447**	0.462**
BMI				1	0.702**	0.172*	0.092	0.662**
Calf circumference					1	0.484**	0.404**	0.781**
Toe grip strength						1	0.601**	0.490**
Handgrip strength							1	0.483**
ASMI								1

TGS (Supplementary Table 1).

3.2. Subgroup results

To explore age-related differences in muscle and anthropometric measures, participants were further stratified into two age groups. Among women aged 50 to 65, the proportion of individuals meeting the criteria for sarcopenia was 3.8 %, compared to 24.2 % in those aged 65 and above. Similarly, the proportions were 1.4 % and 26.4 % in younger and older men, respectively (Table 1, upper panel).

To provide reference values for these subgroups, the lowest quintile (bottom 20 %) values for calf circumference, TGS, HGS, and ASMI were calculated separately by sex and age group. Notably, the lowest quintile thresholds among younger adults were generally higher than those observed in the older group. For example, the lowest quintile for ASMI in younger women was 5.68 kg/m², compared to 5.12 kg/m² in older women; in younger men, it was 7.32 kg/m² versus 6.51 kg/m² in their older counterparts. Similarly, for handgrip strength, the lowest quintile in younger women was 19.5 kg compared to 14.3 kg in older women,

and 32.02 kg in younger men versus 22.04 kg in older men (Table 1, lower panel).

4. Discussion

Due to sex-specific diagnostic thresholds defined by EWGSOP2 and the role of sex as a strong effect modifier, we stratified all analyses by sex. This approach enhances clinical interpretability by allowing clearer identification of predictors relevant to each sex and reflects underlying biological and physiological differences in sarcopenia risk.

Among women, our findings demonstrate that age, height, and calf circumference are strong predictors for identifying sarcopenia, defined as low muscle mass and muscle strength. Calf circumference, closely related to overall muscle mass and lower limb strength, showed high discriminatory power. It reflects the condition of key muscles like the gastrocnemius and soleus [17,18]. These muscles play vital roles in mobility and stability, making calf circumference a practical, non-invasive proxy for muscle integrity [19].

Toe grip strength primarily reflects the function of localized muscles

Table 3

Predictive models for sarcopenia in women (upper panel) and men (lower panel). Abbreviations: AUROC, area under receiver-operating characteristic curve; S.E., standard error of the AUROC.

Women (N = 591)		Odds ratio	95 % confidence interval	p	AUROC	S.E.
Model 1	Toe grip strength	0.761	0.699–0.829	<0.001	0.736	0.025
Model 2	Age	1.069	1.035–1.104	<0.001	0.885	0.017
	Height	0.867	0.823–0.914	<0.001		
	Calf circumference	0.726	0.656–0.805	<0.001		
Model 3	Age	1.056	1.019–1.095	0.003	0.886	0.017
	Height	0.867	0.822–0.914	<0.001		
	Calf circumference	0.730	0.659–0.810	<0.001		
	Toe grip strength	0.923	0.830–1.028	0.144		
Men (N = 180)		Odds ratio	95 % confidence interval	p	AUROC	S.E.
Model 1	Toe grip strength	0.726	0.633–0.833	<0.001	0.825	0.036
Model 2	Age	1.062	0.998–1.130	0.057	0.903	0.026
	Height	0.829	0.737–0.933	0.002		
	Calf circumference	0.692	0.571–0.838	<0.001		
Model 3	Age	1.014	0.949–1.087	0.684	0.924	0.024
	Height	0.801	0.699–0.919	0.001		
	Calf circumference	0.705	0.569–0.874	0.001		
	Toe grip strength	0.783	0.659–0.931	0.006		

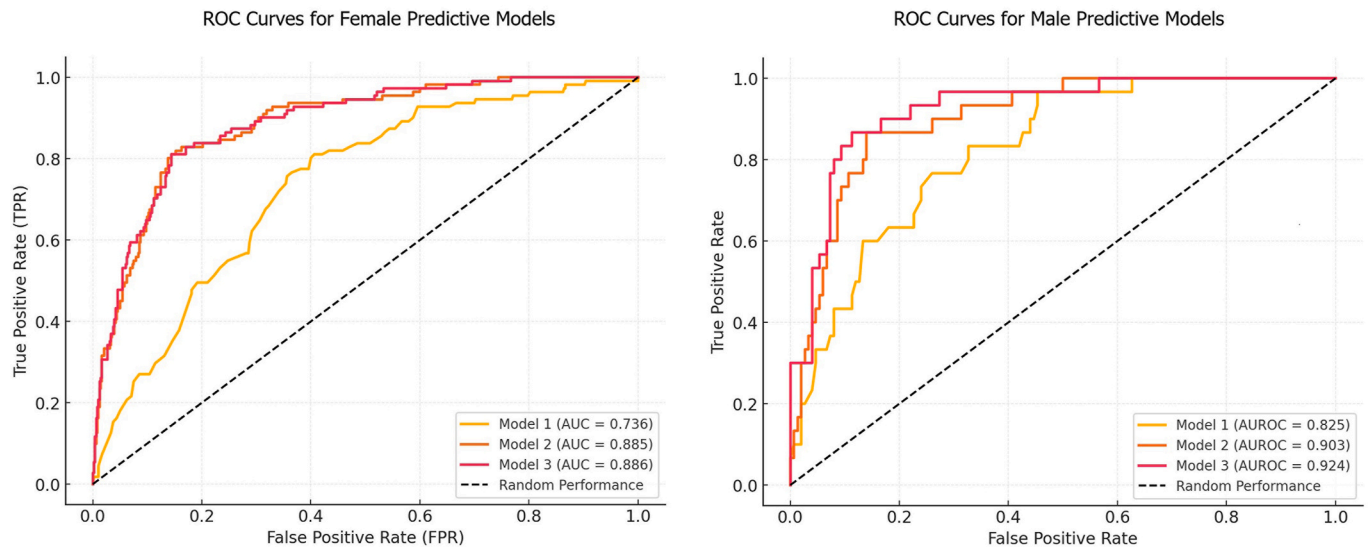


Fig. 1. Receiver operating characteristic (ROC) curves for predictive models in women (left panel) and men (right panel). Each curve represents a different predictive model: Model 1 (toe grip strength alone), Model 2 (age, height, and calf circumference), and Model 3 (Model 2 plus toe grip strength). The area under the ROC curve (AUROC) reflects the discriminatory power of each model. The dashed diagonal line represents the performance of random classification.

in the foot and lower leg, including both intrinsic muscles (e.g., flexor digitorum brevis, lumbricals) and extrinsic muscles (e.g., flexor hallucis longus, flexor digitorum longus) [20]. These muscles are critical for toe flexion and contribute to balance, postural control, and dynamic stability during ambulation [21]. Although TGS reflects aspects of lower-limb muscle function, its contribution to overall muscle mass and systemic strength is limited. In our study, TGS showed a moderate correlation with HGS in both sexes, suggesting shared but distinct functional relevance. However, a direct comparison of predictive ability between TGS and HGS is not appropriate, as HGS is part of the sarcopenia diagnostic criteria and not an independent predictor. TGS did not improve discriminatory power in women but showed potential utility in men. Still, this finding warrants cautious interpretation due to the limited male sample size.

Despite its potential functional relevance, TGS is not recognized as a standard indicator of muscle strength in the context of sarcopenia. Although some population-based studies have proposed reference

values by age and sex, there are no universally accepted or validated cutoff points for clinical use [22]. In our study, we provided sex- and age-specific lowest quintile values for TGS as reference data; however, these are intended for exploratory or comparative purposes only and should not be interpreted as diagnostic thresholds. As such, TGS may at best serve as a supplementary tool in specific contexts, such as when handgrip testing is not feasible, including in individuals with upper limb impairments.

Age was a significant predictor of sarcopenia in women, likely reflecting the natural decline in muscle mass and strength associated with aging [23,24]. In contrast, age was not a significant predictor in men, which may be attributed to their higher baseline muscle mass and greater levels of physical activity, potentially delaying the onset of functional decline [14].

The protective effect of taller height may be partly explained by its association with greater appendicular skeletal muscle mass. In our cohort, height demonstrated a moderate positive correlation with ASMI

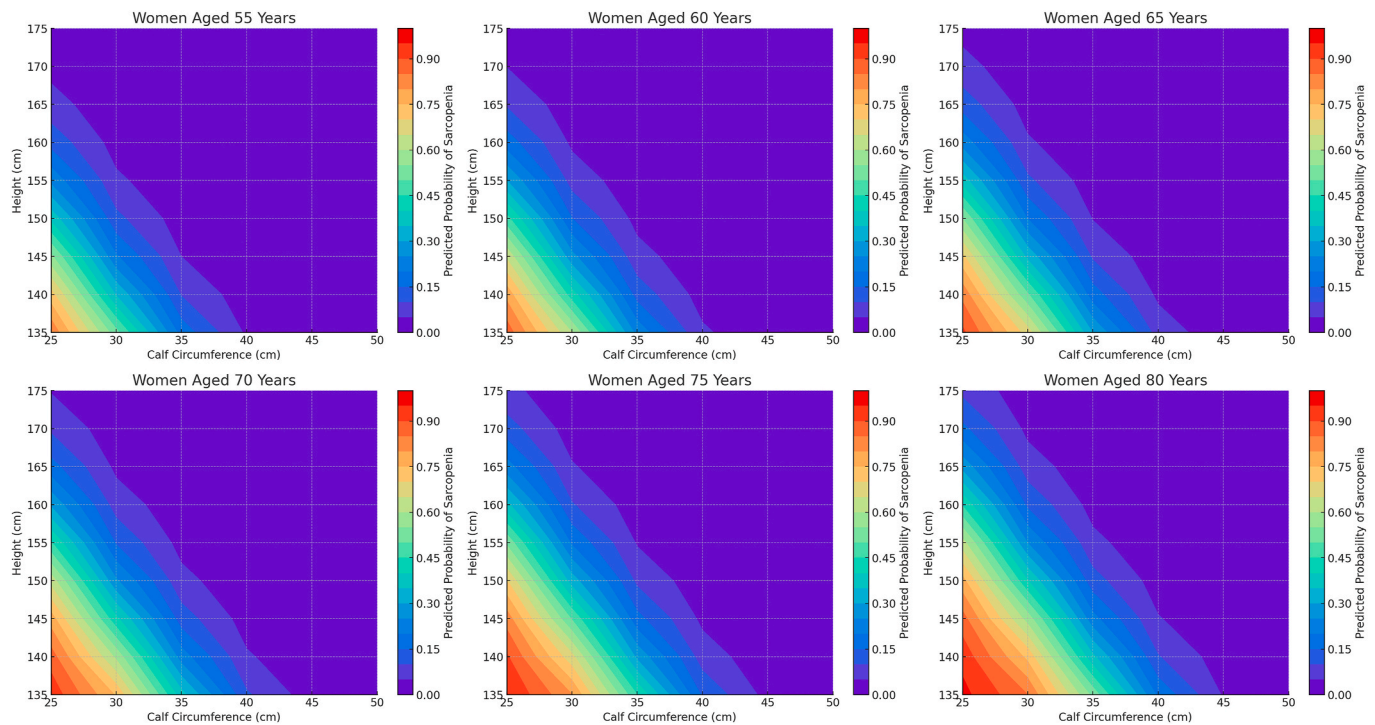


Fig. 2. Color contour plots of predicted probability of sarcopenia in women by age. The x-axis shows calf circumference (cm), and the y-axis shows height (cm). The color gradient represents the estimated probability of sarcopenia, illustrating how the combined effects of calf circumference and height vary across different ages.

in both sexes, suggesting that taller individuals possess larger muscle mass reserves, which may contribute to a lower risk of sarcopenia.

A key limitation of this study was the small number of male participants diagnosed with sarcopenia ($n = 30$), which limited the statistical stability of the models and reduced the power to detect small effects. The events-per-variable (EPV) ratio fell below the recommended threshold of 10, raising concerns about model reliability. Additionally, selection bias may have occurred, as the sample may not represent the broader male population. Cultural norms, lower health awareness, and time constraints likely contributed to reduced male participation in health screening programs [25]. Thus, findings related to TGS in men should be interpreted with caution. Future studies should validate these results in larger, more diverse male cohorts using targeted recruitment strategies such as workplace-based screening and community education efforts [26].

Additionally, the use of convenience sampling from a self-funded health program may have introduced selection bias, favoring individuals who were more health-conscious, health-literate, or of higher socioeconomic status. This could underestimate the true prevalence and variability of sarcopenia in the general population. Those with lower mobility or greater functional decline, who are at higher risk, may have been underrepresented. Future studies may benefit from stratified sampling to improve representativeness. These limitations should be considered when interpreting the findings.

While EWGSOP2 criteria were designed for older adults, applying them to individuals aged 50 to 65 may overlook early muscle decline. Our subgroup analysis showed higher lowest-quintile values in the younger group, suggesting that some cases could be missed using geriatric cutoffs. Age-specific reference standards may help improve early detection in this population.

Finally, while the toe grip dynamometer did not significantly enhance sarcopenia diagnosis in this study, it may hold promise for assessing outcomes like fall risk or balance impairments [27–29]. These applications merit further investigation to determine the tool's broader clinical utility.

In summary, this study highlights the strong discriminatory power of

age, calf circumference, and height for identifying sarcopenia in women, offering a simple and accessible screening approach. While toe grip strength was less effective for diagnosis, it may have specific applications in men or for assessing mobility. Expanding sarcopenia awareness and health literacy beyond hospitals, through community and workplace education tailored to older adults, may improve early recognition and intervention. Future research should validate these findings in diverse populations and explore innovative tools to enhance sarcopenia diagnosis and management.

Authors' contributions

Study concept and design: Hung CC, Fu SH, Cheng BW, Lin JW, Acquisition of data: Chen YJ, Hung CC, Fu SH, Analysis and interpretation of data: Chen YJ, Lin JW, Drafting of the manuscript: Chen YJ, Li AT, Lin JW, Critical revision of the manuscript for important intellectual content: Cheng BW, Li AT, Fu SH, Statistical analysis: Li AT, Lin JW, Study supervision: Hung CC, Cheng BW, Lin JW.

Consent for publication

All authors have read and approved the final manuscript and certify that this work is original, has not been published previously, and is not under consideration for publication elsewhere. In the event of acceptance, all authors agree to transfer copyright ownership.

Availability of data and materials

The datasets generated and/or analyzed during this study are not publicly available due to institutional and governmental restrictions on data sharing. However, all data supporting the conclusions are included within the manuscript.

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Declaration of competing interest

None.

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Abbreviations

ASMI	Appendicular Skeletal Muscle Mass Index
AUROC	Area Under the Receiver Operating Characteristic Curve
AWGS	Asian Working Group for Sarcopenia
BIA	Bioelectrical Impedance Analysis
BMI	Body Mass Index
EPV	Events per Variable
EWGSOP	European Working Group on Sarcopenia in Older People
EWGSOP2	The revised European Working Group on Sarcopenia in Older People
HGS	Handgrip Strength
HMC	Health Management Center
IRB	Institutional Review Board
NTUHYL	National Taiwan University Hospital Yunlin Branch
OR	Odds Ratio
ROC	Receiver Operating Characteristic
SD	Standard Deviation
TGS	Toe Grip Strength

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jfma.2025.06.015>.

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